ZHEYU JIANG

School of Chemical Engineering
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EDUCATION

Ph.D., Purdue University, Chemical Engineering 2014 – 2018

B.Ch.E. (Honors), *magna cum laude*, University of Minnesota 2010 – 2014

PROFESSIONAL APPOINTMENTS

Oklahoma State University August 2021 – Present
Assistant Professor

- Lead the Computational Laboratory of Advanced Manufacturing and Sustainability (CLAMS) at OSU
- Develop mathematical and computational theories and tools for sustainable energy and chemical production, digital twins, precision agriculture, and food-energy-water nexus

The Dow Chemical Company/Corteva Agriscience December 2018 – July 2021
Research Investigator - Process Development Engineer

- Developed robust and low-cost synthetic routes and optimized separation and crystallization technologies to enable the launch of Adavelt fungicide by reducing its solvent consumption by 60% and costs of manufacturing by 40%
- As the company's subject matter expert on liquid separations, designed, optimized, and implemented several new solvent purification and recovery processes for multiple crop protection actives during their process development and manufacturing stages

Purdue University August 2014 – October 2018
Graduate Research Assistant

- Solved a longstanding problem in chemical engineering of developing a shortcut method for minimum reflux calculation for multicomponent distillation in multi-feed, multi-product columns
- Developed the first enumeration based global optimization algorithm to identify distillation configurations that can save up to 50% of total cost or total exergy loss compared to conventional schemes
- For the first time, proposed a systematic multi-layer approach for process intensification in multicomponent distillation, offering industrial practitioners an easy-to-follow recipe to synthesize numerous new and intensified configurations that further enhance operability, improve efficiency, and reduce costs

The Dow Chemical Company May 2016 – August 2016
PhD R&D Intern

- Synthesized novel multicomponent distillation sequences for several Dow's core processes that saved up to 10-15% in energy consumption and capital cost compared to current most effective technologies
- Hosted weekly meetings involving R&D, engineering, and business experts regarding potential innovation and retrofit opportunities for implementing these newly identified configurations

University of Minnesota November 2012 – May 2014
Undergraduate Research Assistant

- Developed the first solution processable method to achieve purification of sub-100 nm thin film of exfoliated MFI zeolite nanosheets using density gradient centrifugation
- Optimized experimental procedure that fabricated high-quality α-alumina membrane supports based on colloidal dispersion processing for MFI nanosheet coating
- Studied complete removal of organic structural directing agent in MFI nanosheets by acid treatment

Honeywell UOP June 2013 – August 2013
Engineering Support Specialist

- Created new UniSim based process simulations for UOP's Oleflex and FCC technologies and successfully built activity coefficient models to describe the VLE and LLE for all major components involved
- Established a crude oil thermodynamic properties databank for major global oil reserves

PROFESSIONAL SERVICE

Technical Committee Member, Process Systems Engineering 2027 Conference 2026 – 2027
Associate Editor, Computational Methods sector, *Frontiers in Chemical Engineering* 2026 – Present
Session Chair and International Scientific Committee, the 10th International Conference on Foundations of Computer-Aided Process Design (FOCAPD) 2024
Guest Editor, Data-Driven Modeling, Optimization and Control of Chemical Processes, *Processes* 2024
Chair/Co-Chair, CAST Division 10a,c,d,e AIChE Annual Meeting 2023 – 2024
Co-chair, Invited Session: Decarbonization of Chemical Process Industries via Electrification, INFORMS Annual Meeting 2024
Chair, Advances in Machine Learning, FOCAPO/CPC Conference 2023 2023
Ad-hoc and panel reviewer, NSF, USDA, USGS 2024 – Present
Reviewer, *AIChE Journal*, *Computers and Chemical Engineering*, *Industrial & Engineering Chemistry Research*, *Chemical Engineering Research and Design*, *ACS Omega*, *Journal of Taiwan Institute of Chemical Engineers*, *Chemical Engineering Transactions*, *Digital Chemical Engineering, Artificial Intelligence Review, Engineering with Computers, IEEE Transactions on Human-Machine Systems* 2021 – Present

SELECTED GRANTS

5. **Single PI**, "Theory Guided, Automated Neural Operator Design for Digital Twin Simulations", *NVIDIA Academic Grant Program for Researchers*, October 2026 – March 2027, 35,000 H100 GPU-Hours ($99,531 equivalent in value)
4. **Single PI**, "CAREER: Data-driven, Physics-augmented Process Systems Engineering Framework for Digital, Sustainable Agriculture", *National Science Foundation* (Award Number: 2442806), January 2025 – December 2030, $500,000
3. **Co-PI**, "EAGER: CET: Decentralized Algorithms for Integrating Decarbonized Chemical Process Heating with Renewable-driven, Electric Power Systems", *National Science Foundation* (Award Number: 2343072), July 2024 – June 2026, $299,050 (Jiang: $152,780)
2. **Single PI**, "FARM: Fast, Accurate, Robust Fault Detection and Diagnosis Software for Industrial Distillation Monitoring", *OCAST Oklahoma Applied Research Support* (Award Number: AR24-069), April 2024 – March 2026, $136,000
1. **Single PI**, "I-Corps: Process Monitoring and Diagnosis Software for Decarbonizing Industrial Distillation", *National Science Foundation* (Award Number: 2331080), June 2023 – November 2024, $50,000

RECOGNITIONS AND AWARDS

NVIDIA Academic Grant Program Award, NVIDIA AI 2026
Professor of the Year Award, Omega Chi Epsilon and School of Chemical Engineering, OSU 2026
Excellent Faculty Award, College of Engineering, Architecture and Technology, OSU 2026
National Science Foundation CAREER Award 2025
Foundations of Process Analytics and Machine Learning 2023 Travel Award for Junior Faculty 2023
Early-career pioneering researcher featured in 2022 Futures Issue, AIChE Journal 2022
People's Choice Award, Corteva Agriscience 2019
AIChE Separations Division Graduate Student Research Award 2018
Eastman Graduate Travel Grant, Purdue University 2017
Purdue Graduate Student Government Travel Grant, Purdue University 2016

Global Excellence Scholarship, UMN 2010 – 2014
College of Science and Engineering Merit Scholarship, UMN 2012
Charles A. Mann Award, Department of Chemical Engineering and Material Science, UMN 2012

PEER-REVIEWED BOOK CHAPTERS (∗ indicates corresponding author)

1. Naraghi SG, Kareck T, Xiao L, Reed R, Ramanan P, Jiang Z. *Decarbonization of Steam Cracking for Clean Olefins Production: Optimal Microgrid Scheduling*, chap. 10, pp. 251–278. John Wiley & Sons, Ltd. 2026;

PEER-REVIEWED JOURNAL PUBLICATIONS (∗ indicates corresponding author)

14. Simra Shoaib S, Hanchey H, Sun W, Jiang, Z*, Tang X. Online Adaptive Model-based Stochastic Control with Limited Data: A Case Study with A Colloidal Self-Assembly System. *AIChE Journal*. 2026;72:e70662
13. Song Z, Jiang, Z*. MP-FVM: Enhancing finite volume method for water infiltration modeling in unsaturated soils via message-passing encoder-decoder network. *Computers and Geotechnics*. 2026; 190:107745
12. Jiang Z*, Tawarmalani M, Agrawal R. Minimum reflux calculation for multicomponent distillation in multi-feed, multi-product columns: Algorithms and examples. *AIChE Journal*. 2025;71(11):e70016
11. Fang W, Ji C, Ma F, Jiang Z, Wang J, Sun W. Harnessing Process Knowledge via Soft Sensing for Accurate Online Monitoring: A Study on the Dow Data Challenge Problem. *Industrial & Engineering Chemistry Research*. 2025;64(31):15363–15376
10. Miraliakbar A, Ma F, Jiang Z*. Online fault detection and classification of chemical process systems leveraging statistical process control and Riemannian geometric analysis. *Computers & Chemical Engineering*. 2025;200:109177.
    • Selected as one of the best contributions from Foundations of Computer Aided Process Design (FOCAPD) 2024
9. Ma F, Ji C, Wang J, Sun W, Tang X, Jiang Z*. MOLA: Enhancing industrial process monitoring using a multi-block orthogonal long short-term memory autoencoder. *Processes*. 2024;12(12):2824.
    • Invited special issue: Process Modeling, Simulation, and Optimization in Chemical Engineering
8. Jiang Z*, Tawarmalani M, Agrawal R. Minimum reflux calculation for multicomponent distillation in multi-feed, multi-product columns: Mathematical model. *AIChE Journal*. 2022;68:e17929.
    • Invited article in the 2022 Futures Issue of AIChE Journal featuring pioneering early career researchers
7. Jiang Z*. A shortcut minimum reflux calculation method for distillation columns separating multicomponent homogeneous azeotropic mixtures. *Le Scientifique*. 2020;2020(1):17–25.
    • Featured in the inaugural issue of *Le Scientifique*, the peer-reviewed academic journal for Corteva scientists
6. Jiang Z, Chen Z, Huff J, Shenvi A, Tawarmalani M, Agrawal R. Global minimization of total exergy loss of multicomponent distillation configurations. *AIChE Journal*. 2019;65(11):e16737
5. Jiang Z, Mathew TJ, Huff J, Nallasivam U, Tawarmalani M, Agrawal R. Global optimization of multicomponent distillation configurations: Global minimization of total cost for multicomponent mixture separations. *Computers & Chemical Engineering*. 2019;126:249–262
4. Jiang Z, Agrawal R. Process intensification in multicomponent distillation: A review of recent advancements. *Chemical Engineering Research and Design*. 2019;147:122–145.
    • Invited review article in the special issue on 11th International Conference on Distillation & Absorption
3. Jiang Z, Madenoor Ramapriya G, Tawarmalani M, Agrawal R. Process intensification in multicomponent distillation. *Chemical Engineering Transactions*. 2018;69:841–846

2. Jiang Z, Madenoor Ramapriya G, Tawarmalani M, Agrawal R. Minimum energy of multicomponent distillation systems using minimum additional heat and mass integration sections. *AIChE Journal*. 2018;64(9):3410–3418

1. Agrawal KV, Topuz B, Jiang Z, Nguenkam K, Elyassi B, Francis LF, Tsapatsis M, Navarro M. Solution-processable exfoliated zeolite nanosheets purified by density gradient centrifugation. *AIChE Journal*. 2013;59(9):3458–3467.
   - Invited article in the special issue of AIChE Journal Founders Tribute to Neal R. Amundson

PEER-REVIEWED CONFERENCE PROCEEDINGS (∗ indicates corresponding author)

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17. Song Z, Jiang Z*. Theory Guided and Interpretable Neural Operator Design for Partial Differential Equation Learning. In: *The Fortieth Annual Conference on Neural Information Processing Systems*. 2026; .
    URL `https://openreview.net/forum?id=ddVTfTf0U6`

16. Song Z, Zhen X, Jiang Z*. Can Large Language Models Design Effective Neural Operators for Solving Partial Differential Equations? In: *Forty-Third International Conference on Machine Learning (ICML) 3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents*. 2026; .
    URL `https://openreview.net/forum?id=6GCIuS0oO0`

15. Song Z, Zhen X, Jiang Z*. Adaptive Fourier Decomposition-guided Neural Operator Design for Inverse PDE Problems. In: *Forty-Second Annual Conference on Uncertainty in Artificial Intelligence (UAI)*. 2026; .
    URL `https://openreview.net/forum?id=W7mL4bR8XY`

14. Song Z, Jiang Z*. Adaptive Mamba Neural Operators. In: *The Fourteenth International Conference on Learning Representations*. 2026;
    - The first ICLR main track publication solely authored by scientists from the State of Oklahoma.

    URL `https://openreview.net/forum?id=OenyzvFZPs`

13. Song Z, Jiang Z*. On Optimization of Poles for Adaptive Fourier Decomposition-Inspired Neural Layers. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026; pp. 4031–4035

12. Ghasemi Naraghi S, Kareck T, Jiang Z*. Multi-objective Optimization of Steam Cracking Microgrid for Clean Olefins Production. *Systems & Control Transactions*. 2025;4:837–843

11. Ghasemi Naraghi S, Jiang Z*. Joint Optimization of Fair Facility Allocation and Robust Inventory Management for Perishable Consumer Products. *Systems & Control Transactions*. 2025;4:882–887.
    - Invited as a keynote talk at ESCAPE 35 Conference

10. Song Z, Jiang Z*. A Novel Bayesian Framework for Inverse Problems in Precision Agriculture. *Systems & Control Transactions*. 2025;4:246–251

9. Song Z, Jiang Z*. A Physics-based, Data-driven Numerical Framework for Anomalous Diffusion of Water in Soil. *Systems & Control Transactions*. 2025;4:2391–2397

8. Miraliakbar A, Jiang Z*. Fast, Accurate, and Robust Fault Detection and Diagnosis of Industrial Processes. *Systems & Control Transactions*. 2024;3:322–329

7. Jiang Z*. Online monitoring and robust, reliable fault detection of chemical process systems. *Computer Aided Chemical Engineering*. 2023;52:1621–1626

6. Ghasemi Naraghi S, Jiang Z*. Stochastic optimization of agrochemical supply chains with risk management. *Computer Aided Chemical Engineering*. 2023;52:3335–3340

5. Song Z, Jiang Z*. A data-driven modeling approach for water flow dynamics in soil. *Computer Aided Chemical Engineering*. 2023;52:819–824

4. Ghasemi Naraghi S, Jiang Z*. Stochastic Optimization of Global Agrochemical Supply Chains with Risk Management. In: *Proceedings of IISE Annual Conference and Expo 2023*. New Orleans, LA. 2023;

3. Xie J, Jiang Z*, Yao B. The Effect of Different Optimization Strategies to Physics-Constrained Deep Learning for Soil Moisture Estimation. In: *Proceedings of IISE Annual Conference and Expo 2023*. New Orleans, LA. 2023;

2. Song Z, Jiang Z*. A data-driven random walk approach for solving water flow dynamics in soil systems. In: *Proceedings of Foundations of Computer-Aided Process Operations and Chemical Process Control Conference*. San Antonio, TX. 2023;
   - One of the 48 papers selected for oral presentation at FOCAPO/CPC 2023

1. Jiang Z*. A shortcut model for multicomponent homogeneous azeotropic distillation. In: *Proceedings of Foundations of Computer-Aided Process Operations and Chemical Process Control Conference*. San Antonio, TX. 2023;

WORKING MANUSCRIPTS ($*$ indicates corresponding author)

5. Song Z, Zhen X, Jiang Z*. From Continuous Manifolds to Arithmetic Lattices: A Diophantine Approach to Neural Operators. 2026

4. Ghasemi Naraghi S, Reed R, Kareck T, Ramanan P, Jiang Z*. Toward Decarbonization of Chemical Manufacturing: Joint Optimization of Unit Commitment and Microgrid Operations. 2026

3. Reed R, Ghasemi Naraghi S, Arman Ahmed K, Kareck T, Jiang Z, Ramanan P. Decentralized operations planning of electrified chemical plants with renewable-driven transmission Systems. 2025

2. Vo HQN, Ghasemi Naraghi S, Arman Ahmed K, Jiang Z, Ramanan P. Decentralized scenario generation for stochastic operations planning of electrified chemical plants with renewable-driven power systems. 2025

1. Ghasemi Naraghi S, Jiang, Z*. Toward equitable food pantry network design: Joint optimization of fair food pantry allocation and robust inventory management. 2025

PRESENTATIONS & TALKS

54. Jiang Z. From Data-driven, Physics-Informed Digital Twin to Scientific Machine Learning: AI-Enabled Systems Engineering for Food, Energy, and Water. 2026. Ingram School of Engineering, Texas State University, San Marcos, TX (Host: Damian Valles)

53. Jiang Z. Toward Sustainable Chemical and Food Productions via Systems Engineering Innovations. 2026. Department of Chemical, Biological and Materials Engineering, University of South Florida, Tampa, FL (Host: Lawrence Stern)

52. Jiang Z. Toward Sustainable Chemical and Food Productions via Systems Engineering Innovations. 2026. Process Systems Engineering (PSE) Seminar Series, Purdue University, West Lafayette, IN (Host: Can Li)

51. Jiang Z. Systems Engineering for Sustainable Food and Chemicals Production. 2025. School of Sustainable Chemical, Biological and Materials Engineering, University of Oklahoma, Norman, OK (Host: Ahmad Al-Douri)

50. Song Z, Jiang Z. A Physics-based, Data-driven Numerical Framework for Anomalous Diffusion of Water in Soil. 2025. 35th European Symposium on Computer Aided Process Engineering (ESCAPE)

49. Song Z, Jiang Z. A Novel Bayesian Framework for Inverse Problems in Precision Agriculture. 2025. 35th European Symposium on Computer Aided Process Engineering (ESCAPE)

48. Ghasemi Naraghi S, Jiang Z. Joint Optimization of Fair Facility Allocation and Robust Inventory Management for Perishable Consumer Products. 2025. 35th European Symposium on Computer Aided Process Engineering (ESCAPE)

47. Ghasemi Naraghi S, Kareck T, Reed R, Ramanan P, Jiang Z. Multi-objective Optimization of Steam Cracking Microgrid for Clean Olefins Production. 2025. 35th European Symposium on Computer Aided Process Engineering (ESCAPE)

46. Vo H, Reed R, Ghasemi Naraghi S, Jiang Z, Ramanan P. Decentralized Importance Sampling in Variational Autoencoders to generate Industrial Carbon Emissions Scenarios. 2025. IISE Annual Conference

45. Reed R, Ghasemi Naraghi S, Jiang Z, Ramanan P. Decentralized Operations Planning of Decarbonized Chemical Plants with Renewable-driven Transmission Systems. 2025. IISE Annual Conference

44. Jiang Z. Toward Sustainable Chemical and Food Production via Systems Engineering Approaches. 2025. Graduate Seminar Series, School of Matter, Transport and Energy, Arizona State University, Tempe, AZ (Host: Bhavik Bakshi)

43. Ghasemi Naraghi S, Kareck T, Jiang Z. Decarbonization of Steam Cracking for Clean Olefins Production: Optimal Microgrid Scheduling. 2024. INFORMS Annual Meeting

42. Reed R, Ramanan P, Ghasemi Naraghi S, Jiang Z. Decentralized Operations Planning of Decarbonized Chemical Plants with Renewable-driven Transmission Systems. 2024. INFORMS Annual Meeting

41. Jiang Z. Decarbonization of Chemical Process Industries via Electrification. 2024. INFORMS Annual Meeting

40. Song Z, Jiang Z. DRW-BO: A Bayesian Framework for Parameter Estimation for Fractional Richards Equation with Applications in Precision Agriculture. 2024. AIChE Annual Meeting

39. Rane A, Zomorodiyan M, Feng Y, Jiang Z, Deep A. Fast, Accurate, and Robust Fault Detection and Diagnosis of Industrial Processes. 2024. AIChE Annual Meeting

38. Miraliakbar A, Ma F, Jiang Z. Online Fault Detection and Diagnosis of Industrial Processes Via Data Augmentation and Integrative Learning of Process Knowledge and Fault Propagation Map. 2024. AIChE Annual Meeting

37. Ma F, Miraliakbar A, Tang X, Jiang Z, Sun W. Fast, Accurate Process Monitoring Based on Multi-Block Mutual Information and Nonparametric Statistical Process Control. 2024. AIChE Annual Meeting

36. Ghasemi Naraghi S, Jiang Z. Joint optimization of fair facility allocation and robust inventory management for perishable consumer products. 2024. AIChE Annual Meeting

35. Miraliakbar A, Jiang Z. Fast, Accurate, and Robust Fault Detection and Diagnosis of Industrial Processes. 2024. Foundations of Computer Aided Process Design (FOCAPD) 2024 Conference

34. Jiang Z. Process Systems Engineering Applications in Sustainable Agriculture and Chemical Manufacturing. 2024. PSE Seminar Series, Department of Chemical Engineering and Materials Science, University of Minnesota, Minneapolis, MN (Host: Qi Zhang)

33. Jiang Z. Toward Sustainable, Intensified Food and Chemical Productions via Process Systems Engineering Approaches. 2024. School of Chemical, Biological and Environmental Engineering, Oregon State University, Corvallis, OR (Host: Cory Simon)

32. Jiang Z. Decarbonization of Industrial Distillation via Systems Engineering Approaches. 2023. 2023 ChemE Show, Galveston, TX (Host: Tony Cai)

31. Song Z, Jiang Z. A Computationally Efficient Data-Driven Framework for Solving Water Flow Dynamics in Soil Via Fractional Diffusion Model. 2023. AIChE Annual Meeting

30. Song Z, Jiang Z. A Data-Driven Numerical Method for Solving Water Flow Dynamics in Soil. 2023. AIChE Annual Meeting

29. Song Z, Jiang Z. A Data-Driven Numerical Framework for the Richards Equation for Sustainable Irrigation and Food Production. 2023. AIChE Annual Meeting

28. Miraliakbar A, Jiang Z. Online Monitoring and Robust, Reliable Fault Detection and Diagnosis of Chemical Process Systems. 2023. AIChE Annual Meeting

27. Ghasemi Naraghi S, Jiang Z. Perspective Reformulation of Stochastic Agrochemical Supply Chain Optimization Problem with Mean-Variance Risk Management. 2023. AIChE Annual Meeting

26. Ghasemi Naraghi S, Jiang Z. Stochastic Bilevel Optimization of Agrochemical Supply Chains with Mean-Variance Risk Management. 2023. AIChE Annual Meeting

25. Jiang Z. Creating a Sustainable Food Future via Systems Engineering Approaches. 2023. ACS Southwest Regional Meeting (SWRM), Oklahoma City, OK (Host: H. N. Cheng)

24. Jiang Z. Toward Sustainable Food and Chemical Productions via Systems Engineering Approaches. 2023. Cornell University, Ithaca, NY (Host: Fengqi You)
    - One of the prestigious Ezra's Systems Seminar Series featured by Cornell Systems Engineering Program

23. Jiang Z. Decarbonization of Industrial Distillation via Shortcut modeling, Global Optimization, and Process Intensification – Recent Advancements, Case Studies, and Future Perspectives. 2023. Phillips 66 Research Center, Bartlesville, OK (Host: Karthik Marimuthu)
    - Inaugural quarterly seminar series in 2023 for AIChE Bartlesville Chapter

22. Miraliakbar A, Jiang Z. FARM: A Fast, Accurate, Robust fault detection and diagnosis framework for industrial process Monitoring. 2023. Foundations of Process/product Analytics and Machine learning, Davis, CA

21. Jiang Z. Creating a Sustainable Food Future in the 21st Century. 2023. Department of Biosystems & Agricultural Engineering, Oklahoma State University, Stillwater, OK

20. Jiang Z. AI Applications in Chemical Process Industry. 2023. AIChE Spring Meeting, Houston, TX (Host: Senthil Krishnamoorthy)

19. Song Z, Jiang Z. A data-driven random walk approach for solving water flow dynamics in soil systems. 2023. The Foundations of Computer-Aided Process Operations and Chemical Process Control (FOCAPO/CPC) 2023 Conference, San Antonio, TX

18. Jiang Z. A shortcut model for multicomponent homogeneous azeotropic distillation. 2023. The Foundations of Computer-Aided Process Operations and Chemical Process Control (FOCAPO/CPC) 2023 Conference, San Antonio, TX

17. Jiang Z. Transforming Junior-year Separations Course into an Early-capstone Learning Experience. 2022. ASEE/AIChE Summer School for Engineering Faculty, Golden, CO

16. Jiang Z. Creating a Sustainable Manufacturing and Food Future via Process Systems Engineering Innovations. 2021. School of Industrial Engineering and Management, Oklahoma State University, Stillwater, OK (Host: Joseph Nuamah)

15. Jiang Z. Advancing Future-generation Separation Technologies via Systems Engineering Innovations: Multicomponent Distillation and Beyond. 2021. School of Chemical Engineering, Oklahoma State University, Stillwater, OK

14. Jiang Z. Creating a Sustainable Food Future by 2050 via Systems Engineering Innovations. 2021. Chemical and Biological Engineering Department, University of Wisconsin, Madison, WI (Host: Victor Zavala)

13. Jiang Z. Advancing Future-generation Separation Technologies via Process Systems Engineering Innovations. 2021. Chemical and Biological Engineering Department, University of Wisconsin, Madison, WI (Host: Victor Zavala)
12. Jiang Z. Innovating Future-generation Separation Technologies via Process Systems Engineering. 2020. Virtual AIChE Annual Meeting
11. Jiang Z. Innovating Future-generation Separation Processes via Systems Engineering. 2020. Prof. Fengqi You's group, Cornell University, Ithaca, NY (Host: Fengqi You)
10. Jiang Z. Minimum reflux calculation for multicomponent azeotropic distillation using shortcut method. 2019. AIChE Annual Meeting, Orlando, FL
9. Jiang Z. A shortcut model for multicomponent azeotropic distillation column design. 2019. AIChE Annual Meeting, Orlando, FL
8. Jiang Z. A Modeling Approach to Designing Effective Solvent Exchange and Recycle Processes for Agrochemical Active Ingredient Manufacturing. 2019. Crop Protection Product Design & Process Summit, Indianapolis, IN (Host: Abraham Schuitman)
7. Jiang Z, Tawarmalani M, Agrawal R. Minimum reflux behavior of multicomponent mixture separation using complex distillation columns. 2018. AIChE Annual Meeting, Pittsburg, PA
6. Jiang Z, Tawarmalani M, Agrawal R. A new minimum reflux calculation method for multiple-feed distillation columns distilling ideal multicomponent mixtures. 2017. AIChE Annual Meeting, Minneapolis, MN
5. Jiang Z, Tawarmalani M, Agrawal R. Process intensification in multicomponent distillation. 2017. AIChE Annual Meeting, Minneapolis, MN
4. Jiang Z, Tawarmalani M, Agrawal R. Process intensification in multicomponent distillation. 2017. AIChE Spring Meeting, San Antonio, TX
3. Jiang Z, Madenoor Ramapriya G, Tumbalam Gooty R, Tawarmalani M, Agrawal R. Minimum energy of multicomponent distillation systems using minimum additional number of heat and mass integration sections. 2016. AIChE Annual Meeting, San Francisco, CA
2. Jiang Z, Madenoor Ramapriya G, Tumbalam Gooty R, Tawarmalani M, Agrawal R. Process intensification of multicomponent distillation configurations using minimum additional number of heat and mass integration sections. 2016. AIChE Annual Meeting, San Francisco, CA
1. Jiang Z, Madenoor Ramapriya G, Tumbalam Gooty R, Tawarmalani M, Agrawal R. A method for minimization of total exergy loss over the complete search space of regular distillation configurations. 2016. AIChE Annual Meeting, San Francisco, CA

STUDENT MENTORSHIP

Graduate Students: Zeyuan Song (Spring 22 – Fall 25, now a Postdoctoral Researcher in Department of Mathematics at Lehigh University), Saba Ghasemi Naraghi (Spring 22 – Fall 25, now a Research Scientist at American Airlines), Oluwatimileyin Ogunsola (Spring 26 – Present), Xiaoqun Li (Fall 26 – Present)

Undergraduate Students: Tylee Kareck (Fall 23 – Spring 25, now a Ph.D. student at Texas A&M, NSF GRFP Honorable Mention, CEAT Undergraduate Research Scholarship, Outstanding Chemical Engineering Senior Award, R.N. Maddox Chemical Engineering Design Contest Award), Cole Elder (Fall 25 – Present, serving the U.S. Air Force), Chris Turner (Spring 26 – Present, Wentz Research Scholarship)

DISSERTATION COMMITTEE

Suhao Chen (now Assistant Professor in Industrial Engineering at South Dakota Mines) Summer 22
Batuhan Bal (Chemical Engineering Ph.D. proposal & thesis defense committee) Fall 23, Spring 24

Rashed Islam (Chemical Engineering M.S. thesis defense committee)	Fall 23
Ted Sperry (Chemical Engineering Ph.D. proposal & thesis defense committee)	Spring 24, Spring 25
Ricky Reed (Industrial Engineering and Management M.S. thesis defense committee)	Spring 25
Katrine Hareland (Petroleum Engineering Ph.D. proposal & thesis defense committee)	Fall 25, Spring 26
Kazi Arman Ahmed (Industrial Engineering and Management M.S. thesis defense committee)	Spring 26
Ziying Wang (Electrical Engineering Ph.D. proposal & thesis defense committee)	Spring 26
Omar Hamdy (Petroleum Engineering Ph.D. proposal defense committee)	Fall 26

TEACHING (student course evaluation results are based on a total score of 5)

Lead Instructor, CHE 3113 – Rate Operations II (4.0, 4.4, 4.5, 4.6, 4.8)	Spring 22, 23, 24, 25, 26
Lead Instructor, CHE 4990 – Modern Energy Systems Analysis (4.5)	Fall 25
Lead Instructor, CHE 6010 – CHE Graduate Seminar (4.5, 4.5)	Fall 23, Spring 24
Lead Instructor, CHE 2581 – CHE Sophomore Seminar (4.7, 4.7)	Fall 22, 24
Lead/Co-Instructor, CHE 4124 – Chemical Engineering Design (4.5, 4.4, 4.7)	Fall 21, 23, 25

PROFESSIONAL SOCIETIES AND MEMBERSHIP

Sigma Xi Scientific Research Honor Society, American Institute of Chemical Engineers (AIChE), American Chemical Society (ACS), Institute of Industrial and Systems Engineers (IISE), Institute for Operations Research and the Management Sciences (INFORMS), Institute of Electrical and Electronics Engineers (IEEE)